

Fermion models in two-dimensional curved space-time

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The Thirring and Schwinger models in an arbitrary differentiable two-dimensional space-time are studied. Using a path-integral approach, an analytical solution of the former is obtained, while relevant properties can be discovered from the Schwinger-model results.

Nowadays there is much interest in the study of interacting field theories in curved space-time, for example, in connection with the history of the early universe.¹ Few solutions, if any, are known in four-dimensional space-time, and so the study of two-dimensional models—soluble in flat space-time—is desirable, since it may shed light on relevant properties to be welcomed in more physical theories.

The purpose of this work is to discuss two such models, namely, the Thirring and Schwinger models, using a powerful functional-integral approach recently developed²⁻⁶ after Fujikawa's⁷ observation on the noninvariance of the path-integral measure under chiral transformations.

Our method is a sort of path-integral version of the bosonization technique, but in it there is no need of proposing an explicit fermion parametrization in terms of boson fields (bosonization) as is the case in the operator fit. It explicitly exploits the chiral anomaly in order to decouple fermions from vector fields (either gauge or auxiliary fields) by making a chiral change in the fermion variables. It has the advantage of its simplicity and applicability to theories in which the operator approach presents difficulties (for example, when there are symmetries not preserved by bosonization, as in the two-dimensional QCD model).

Owing to the fact that in our treatment the fermion degrees of freedom are kept, the analysis of correlation functions becomes direct and simple. In this way, exact solutions of various models in flat space-time were obtained economically, and also new insights into the still unsolved cases (such as QCD₂ or chiral Gross-Neveu models) were obtained. As we shall see, we shall obtain the complete solution of the Thirring and Schwinger models for an arbitrary differentiable two-dimensional space-time. For the former, an analytical solution is obtained, while relevant properties can be discovered for the Schwinger-model case.

We shall first consider the Thirring model in two-dimensional curved space-time. This model was already investigated by Birrel and Davies⁸ who gave an exact operator solution obtained from Klaiber's flat-space solution by exploiting conformal invariance. We think it is worthwhile to reobtain the solution using the path-integral approach since our treatment is completely different from the operator fit, yielding to the correlation function in a very simple way, not only for the Thirring

model (where conformal invariance simplifies the solution) but for more involved models, such as, for example, the Schwinger model, which we shall also solve in this work.

The dynamics of the Thirring model in curved space-time is determined by the Lagrangian

$$\mathcal{L}_{\text{Th}} = -\bar{\psi}\gamma^\mu(x)i[\partial_\mu - \Gamma_\mu(x)]\psi - \frac{1}{2}e^2[\bar{\psi}\gamma_\mu(x)\psi][\bar{\psi}\gamma^\mu(x)\psi]. \quad (1)$$

Here, the curved-space-time γ matrices are denoted by $\gamma_\mu(x)$ and satisfy

$$\gamma_\mu(x)\gamma_\nu(x) + \gamma_\nu(x)\gamma_\mu(x) = 2g_{\mu\nu}(x). \quad (2)$$

These matrices are related to the flat-space ones by means of the vierbeins:

$$\gamma_\mu(x) = b_\mu^a(x)\gamma_a, \quad (3)$$

where

$$g_{\mu\nu}(x) = b_\mu^a(x)b_\nu^b(x)\eta_{ab}. \quad (4)$$

The locally flat space-time will be continued to Euclidean space, i.e.,

$$\eta_{ab} = \delta_{ab}. \quad (5)$$

The spinorial affine connection is defined by

$$\Gamma_\mu(x) = \frac{1}{4}\gamma_a\gamma_b b_\lambda^a(x)g^{\lambda\sigma}(x)\nabla_\mu b_\sigma^b(x), \quad (6)$$

with

$$\nabla_\mu b_\sigma^a = \partial_\mu b_\sigma^a - \Gamma_{\mu\sigma}^\lambda b_\lambda^a, \quad (7)$$

where $\Gamma_{\mu\sigma}^\lambda$ is the usual Christoffel symbol. Finally $\bar{\psi}$ is defined as

$$\bar{\psi} = \psi^\dagger\gamma_0. \quad (8)$$

The generating functional for the Green's functions is given by

$$Z(\bar{\theta}, \theta) = \mathcal{N} \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp \left[- \int (\mathcal{L}_{\text{Th}} + \bar{\theta}\psi + \bar{\psi}\theta) d\mu_x \right], \quad (9)$$

where

$$d\mu_x = \sqrt{g} d^2x, \quad (10)$$

$$\sqrt{g} = [\det(g_{\mu\nu})]^{1/2} = \det[b_\mu^a(x)].$$

We now introduce an auxiliary vector field A_μ through the well-known Gaussian identity:

$$\exp \left[\frac{1}{2} e^2 \int (\bar{\psi} \gamma_\mu \psi) (\bar{\psi} \gamma^\mu \psi) d\mu_x \right] = \int \mathcal{D} A_\mu \exp \left[- \int \left(\frac{1}{2} A_\mu A^\mu - e \bar{\psi} \gamma_\mu A^\mu \psi \right) d\mu_x \right]. \quad (11)$$

Inserting expression (11) in Eq. (9) we get

$$Z(\bar{\theta}, \theta) = \int \mathcal{D} \bar{\psi} \mathcal{D} \psi \mathcal{D} A_\mu \exp \left[- \int \mathcal{L} d\mu_x + \text{source terms} \right], \quad (12)$$

where

$$\mathcal{L} = -\bar{\psi} [i \gamma^\mu(x) (\partial_\mu - \Gamma_\mu) + e \gamma_\mu A^\mu] \psi + \frac{1}{2} A_\mu A^\mu. \quad (13)$$

We are then faced with an effective Lagrangian where fermions are coupled canonically to a vector field A_μ carrying mass dimensions. The Lagrangian $\mathcal{L}$ has no A_μ -kinetic term and does not correspond to a gauge field theory: the vector field represents two degrees of freedom. Now, A_μ can be completely decoupled from the fermions by making the following change of variables:

$$\begin{aligned} \psi(x) &= e^{\gamma_5 \phi(x) + i \eta(x)} \chi(x), \\ \bar{\psi}(x) &= \bar{\chi}(x) e^{\gamma_5 \phi(x) - i \eta(x)}, \end{aligned} \quad (14)$$

$$A^\mu(x) = -\frac{1}{e} \left[\frac{\epsilon^{\mu\nu}}{\sqrt{g}} \partial_\nu \phi - \partial^\mu \eta \right].$$

Here γ_5 is the usual flat-space γ_5 matrix,

$$\gamma_5 = i \gamma_0 \gamma_1 \quad \text{and} \quad \gamma_5^2 = 1, \quad (15)$$

satisfying

$$\gamma^a \gamma_5 = i \epsilon^{ab} \gamma_b \quad (\epsilon^{01} = -\epsilon^{10} = 1). \quad (16)$$

In curved space-time, this relation takes the form

$$\gamma^\mu(x) \gamma_5 = i \frac{\epsilon^{\mu\nu}}{\sqrt{g}} \gamma_\nu(x). \quad (17)$$

This is precisely the property that allows the fermion decoupling as happens in the flat-space case (see, for example, Ref. 2). In terms of the new variables χ , ϕ , and η , the Lagrangian reads

$$\begin{aligned} \mathcal{L} &= -\bar{\chi} \gamma^\mu(x) i (\partial_\mu - \Gamma_\mu) \chi + \frac{1}{2e^2} (\partial_\mu \phi)(\partial^\mu \phi) \\ &+ \frac{1}{2e^2} (\partial_\mu \eta)(\partial^\mu \eta) + \frac{1}{e^2 \sqrt{g}} \epsilon^{\mu\nu} (\partial_\mu \phi)(\partial_\nu \eta). \end{aligned} \quad (18)$$

Now, it is well established that the transformation (14) is not free of cost at the quantum level. Indeed, the path-integral measure is not invariant under the chiral part of the fermionic transformation,⁷ and this fact has to be taken into account in order to write the generating functional in terms of the new fields.

Concerning the vector field, it is easy to see that the transformation (14) gives rise to the Jacobian

$$\mathcal{D} A^0 \mathcal{D} A^1 = \det \left[-\frac{1}{e^2} \frac{1}{\sqrt{g}} (\partial_1 \partial^1 + \partial_0 \partial^0) \right] \mathcal{D} \phi \mathcal{D} \eta. \quad (19)$$

The determinant in Eq. (19) is curvature dependent. However, since in our analysis the metric is not quantized, it can be absorbed in the normalization factor since it cancels out when computing properly normalized correlation functions.

Since the fermionic measure can be defined to be invariant under local U(1) rotations, it is the chiral part of the fermionic transformation that will give rise to a nontrivial Jacobian. Its contribution can be evaluated using the method developed in Refs. 2–6. We shall only sketch the principal steps of this computation, the details having already been presented in Ref. 6.

We start from a chiral transformation depending on a real parameter t :

$$\begin{aligned} \psi_t &= e^{-\gamma_5 \phi t} \psi, \\ \bar{\psi}_t &= \bar{\psi} e^{-\gamma_5 \phi t}. \end{aligned} \quad (20)$$

The finite chiral part of transformation (14) will be built up by varying t from 0 to 1.

Consider the fermionic path integral

$$Z_F = \int \mathcal{D} \bar{\psi} \mathcal{D} \psi e^{-\int \bar{\psi} \mathcal{D} \psi d\mu_x} = \det \mathcal{D}, \quad (21)$$

where $\mathcal{D}$ is the Dirac operator

$$i \gamma_\mu(x) (\partial^\mu - \Gamma^\mu - i e A^\mu).$$

After the change (20), Z_F reads

$$Z_F = \int \mathcal{D} \bar{\psi}_t \mathcal{D} \psi_t J(t) e^{-\int \bar{\psi}_t \mathcal{D}_t \psi_t d\mu_x} = J(t) \det \mathcal{D}_t, \quad (22)$$

where

$$\mathcal{D}_t = e^{\gamma_5 \phi t} \mathcal{D} e^{\gamma_5 \phi t} \quad (23)$$

and $J(t)$ is the Jacobian associated with the chiral transformation (20). The Jacobian we are looking for is just $J(t)|_{t=1}$. Since Z_F is t independent, we can write

$$\frac{dZ_F}{dt} = 0 = \frac{d}{dt} [J(t) \det \mathcal{D}_t] \quad (24)$$

or, after integration,

$$J = J(1) = \exp \left[- \int_0^1 W'(t) dt \right] \quad (25)$$

with

$$W(t) = \ln \det \mathcal{D}_t. \quad (26)$$

Now, it is important to stress that all determinants appearing above only make sense after a regularization has been introduced. A particularly suitable regularization is provided by the ζ -function method⁹ which we shall adopt. In this approach, $W(t)$ reads

$$W(t) = - \frac{d}{ds} \zeta(s, \mathcal{D}_t) \Big|_{s=0}. \quad (27)$$

Here $\zeta(s, A)$ can be defined from the analytic extension of the series

$$\xi(s, A) = \sum \lambda_A^{-s} \quad (28)$$

which converges for sufficiently large Res . It has been proven in Ref. 10 that, given a general invertible elliptic operator D defined on a compact manifold without boundary, the ξ function can be extended to a meromorphic function in the whole complex plane s , regular at $s=0$. (From here on we shall consider our model either on a compact space or on a suitable compactified one. We shall also assume that the invertibility condition is satisfied; however, zero modes can be considered following the approach developed in Ref. 12.) It can be also written in the form

$$\xi(s, D_t) = \text{Tr}[D_t^{-s}] = \int \text{tr}[K_{-s}(D_t; x, x)] d\mu_x. \quad (29)$$

The function $K_{-s}(D_t; x, x)$ is the meromorphic extension of the evaluation of the kernel of D_t^{-s} . In (29), we denote by tr the matricial trace.

With our choice of Dirac matrices D_t takes the form

$$D_t = i\gamma_\mu(x)(\partial^\mu - \Gamma^\mu) + e(1-t)\gamma_\mu(x)A^\mu \quad (30)$$

and then¹¹

$$\begin{aligned} \frac{dW(t)}{dt} &= - \lim_{\Delta \rightarrow 0} \frac{1}{\Delta t} \left[\frac{d\xi}{ds}(s, D_{t+\Delta t}) - \frac{d\xi}{ds}(s, D_t) \right] \Big|_{s=0} \\ &= 2 \text{Tr}[D_t^{-s} \gamma_5 \phi] \end{aligned} \quad (31)$$

and finally,

$$\ln J = -2 \int d\mu_x \int_0^1 dt \text{tr}[K_0(D_t; x, x) \gamma_5] \phi. \quad (32)$$

The kernel K_0 can be evaluated using Seeley's coefficients as explained in Ref. 6. The answer is

$$\text{tr}[K_0(D_t; x, x) \gamma_5] = -\frac{e}{2\pi} (1-t) \frac{\epsilon^{\mu\nu}}{\sqrt{g}} F_{\mu\nu}, \quad (33)$$

and hence the final result for the Jacobian reads

$$\ln J = \frac{e}{4\pi} \int \epsilon^{\mu\nu} F_{\mu\nu} \frac{1}{\sqrt{g}} \phi d\mu_x \quad (34)$$

or

$$\ln J = -\frac{1}{2\pi} \int d\mu_x (\partial_\mu \phi)(\partial^\mu \phi). \quad (35)$$

Notice that Eq. (33) has no curvature term, in contrast with the four-dimensional case.¹³ The complete generating functional Z , Eq. (12), can then be written as

$$\begin{aligned} Z[\bar{\theta}, \theta] &= \mathcal{N} \int \mathcal{D}\bar{\chi} \mathcal{D}\chi \mathcal{D}\phi \mathcal{D}\eta \\ &\quad \times \exp \left[- \int (\mathcal{L}_{\text{eff}} + \bar{\theta} e^{\gamma_5 \phi + i\eta} \chi \right. \\ &\quad \left. + \bar{\chi} e^{\gamma_5 \phi - i\eta} \theta) d\mu_x \right] \end{aligned} \quad (36)$$

with $\mathcal{L}_{\text{eff}}$ given by

$$\begin{aligned} \mathcal{L}_{\text{eff}} &= -\bar{\chi}_i \gamma_\mu (\partial^\mu - \Gamma^\mu) \bar{\chi} + \frac{1}{2e^2} \left[1 + \frac{e^2}{\pi} \right] (\partial_\mu \phi)(\partial^\mu \phi) \\ &\quad + \frac{1}{2e^2} (\partial_\mu \eta)(\partial^\mu \eta). \end{aligned} \quad (37)$$

A very simple effective Lagrangian is then obtained. It corresponds to two massless free scalar fields completely decoupled (except for the source terms) from the fermion field χ . It is now straightforward to compute any fermion correlation function. For example, the two-point function can be obtained from

$$\begin{aligned} \langle \psi(x) \bar{\psi}(x') \rangle &= \frac{1}{Z} \frac{\delta^2 Z}{\delta \bar{\theta}(x) \delta \theta(x')} \Big|_{\bar{\theta}=\theta=0} \\ &= \langle \chi(x) \bar{\chi}(x') \rangle_0 \langle \exp \{ i[\eta(x) - \eta(x')] \\ &\quad + \gamma_5 [\phi(x) - \phi(x')] \} \rangle. \end{aligned} \quad (38)$$

Now,

$$G_0(x, x') = \langle \chi(x) \bar{\chi}(x') \rangle_0 \quad (39)$$

is the Green's function of a massless free fermion and hence it can be obtained from the flat one:

$$G_0^{\text{flat}}(x - x') = \text{const} \frac{\gamma_a (x^a - x'^a)}{(x - x')^2} \quad (40)$$

by a conformal transformation. In fact, it is well known that in any two-dimensional manifold the metric can be chosen in a conformally flat form:

$$g^{\mu\nu}(x) = e^{-2\Sigma(x)} \eta^{ab} \delta_a^\mu \delta_b^\nu. \quad (41)$$

In terms of Σ , G_0 can be written as

$$G_0(x, x') = e^{-\Sigma(x)/2} G_0^{\text{flat}}(x - x') e^{-\Sigma(x')/2}. \quad (42)$$

Concerning the scalar-field factor, it can be expressed as

$$\begin{aligned} \langle \exp \{ i[\eta(x) - \eta(x')] + \gamma_5 [\phi(x) - \phi(x')] \} \rangle \\ = \exp \left[\left[\frac{1}{\alpha^2} - \frac{1}{\beta^2} \right] \Delta_F(x, x') \right], \end{aligned} \quad (43)$$

where

$$\alpha^2 = \frac{1}{g^2} \quad \text{and} \quad \beta^2 = \frac{1}{g^2} + \frac{1}{\pi}, \quad (44)$$

and Δ_F is the free massless scalar propagator in curved space-time, satisfying

$$\square \Delta_F(x, x') \equiv \frac{1}{\sqrt{g}} \partial_\mu [\sqrt{g} \partial^\mu \Delta_F(x, x')] = \frac{1}{\sqrt{g}} \delta(x - x'). \quad (45)$$

Note that Eq. (45) has the flat-space form. For an asymptotically flat space we have

$$\Delta_F(z) = -\frac{1}{4\pi} \ln(\mu^2 z_\mu z^\mu) \quad (46)$$

and finally

$$\begin{aligned} \langle \psi(x) \bar{\psi}(x') \rangle &= \text{const } e^{-\Sigma(x)/2} \frac{\gamma_a z^a}{|z_\mu z^\mu|} \\ &\times (\mu^2 |z_\mu z^\mu|)^{-(1/4\pi)(1/\alpha^2 - 1/\beta^2)} \\ &\times e^{-1\Sigma(x')/2}, \end{aligned} \quad (47)$$

where

$$z_\mu = x_\mu - x'_\mu.$$

As we stated above, this correlation function has been already obtained by Birrel and Davies⁸ by transforming the flat-space Klaiber solution¹⁴ to curved space. It is worthwhile noting that the Jacobian contribution prevents (47) from being just the free fermion Green's function (since $\alpha^2 \neq \beta^2$ due to its presence). One can easily extend this result to a general n -point correlation function, as well as to any matrix element of composite fermion fields relevant for the computation of structure functions.

The analysis of the Schwinger model parallels the one discussed above except for the fact that in the former A_μ is a gauge field and, hence, not only does one have a kinetic term in the Lagrangian but also the necessity of fixing the gauge. The Lagrangian for this model reads

$$\mathcal{L}_{\text{sch}} = -\bar{\psi} \gamma_\mu (i\partial^\mu - i\Gamma^\mu + eA^\mu) \psi + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \mathcal{L}_{\text{gf}}, \quad (48)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and $\mathcal{L}_{\text{gf}}$ corresponds to the gauge-fixing term. Although the computations can be performed in a general gauge, we consider for simplicity the Lorentz gauge, i.e., $\partial_\mu A^\mu = 0$ and hence

$$\mathcal{L}_{\text{gf}} = \frac{1}{2\alpha} (\partial_\mu A^\mu)^2.$$

The decoupling change of variables reads in this case [cf. Eq. (14)]

$$\begin{aligned} \psi(x) &= e^{\gamma_s \phi(x)} \chi(x), \\ \bar{\psi}(x) &= \bar{\chi}(x) e^{\gamma_s \phi(x)}, \\ A^\mu(x) &= -\frac{1}{e} \left[\frac{1}{\sqrt{g}} \epsilon^{\mu\nu} \partial_\nu \phi \right]. \end{aligned} \quad (49)$$

In terms of the new variables, Lagrangian (48) reads

$$\mathcal{L}_{\text{sch}}^{\text{eff}} = -i\bar{\chi} \gamma_\mu (\partial^\mu - \Gamma^\mu) \chi + \frac{1}{2e^2} \phi \square \phi - \frac{1}{2\pi} \phi \square \phi. \quad (50)$$

In Eq. (50) we have included as before the chiral Jacobian arising from the transformation (49). (This Jacobian coincides with the previously obtained Thirring-model Jacobian.) The generating functional for fermion Green's functions now reads

$$\begin{aligned} Z[\bar{\theta}, \theta] &= \int \mathcal{D}\bar{\chi} \mathcal{D}\chi \mathcal{D}\phi \\ &\times \exp \left[- \int (\mathcal{L}_{\text{sch}}^{\text{eff}} + \bar{\theta} e^{\gamma_s \phi} \chi \right. \\ &\quad \left. + \bar{\chi} e^{\gamma_s \phi} \theta) d\mu_x \right]. \end{aligned} \quad (51)$$

Any Green's function can be easily obtained from Eq. (51). For example, the two-point function is given by

$$\begin{aligned} \langle \psi(x) \bar{\psi}(x') \rangle &= \frac{1}{Z} \frac{\delta^2 Z[\bar{\theta}, \theta]}{\delta \bar{\theta}(x) \delta \theta(x')} \Big|_{\bar{\theta}=\theta=0} \\ &= G_0(x, x') e^{-\Delta(x, x')}, \end{aligned} \quad (52)$$

where G_0 is the free fermion Green's function (42) and Δ is the scalar propagator satisfying

$$\left[\frac{1}{e^2} \square \square - \frac{1}{\pi} \square \right] \Delta(x) = \frac{1}{\sqrt{g}} \delta(x). \quad (53)$$

Now one can easily show that

$$\Delta(x) = \pi \left[\Delta_F \left[\frac{e^2}{\pi}, x \right] - \Delta_F(0, x) \right],$$

where

$$\begin{aligned} (\square - m^2) \Delta_F(m^2, x) &= \frac{1}{\sqrt{g}} \delta(x), \\ \square \Delta_F(0, x) &= \frac{1}{\sqrt{g}} \delta(x). \end{aligned} \quad (54)$$

Hence, apart from the free fermion sector, there is a scalar massive particle, with $\Delta_F(0, x)$ the manifestation of a massless (gauge) excitation. This last contribution can be explicitly computed and corresponds to (46). On the other hand, $\Delta_F(m^2, x)$ cannot be, in general, evaluated in an exact form. In certain cases, however, its behavior for large and small distances can be analyzed.

Indeed, for small x ,

$$\Delta_F(m^2, x) - \Delta_F(0, x) \rightarrow 0$$

and the fermions are free at short distances. For large distances and asymptotically flat space-time,

$$\Delta_F \left[\frac{e^2}{\pi}, x, x' \right] \rightarrow \Delta_F^{\text{flat}} \left[\frac{e^2}{\pi}, x, x' \right]$$

and the analysis parallels that of the flat Schwinger model. As in that case, as is well known, one has to use the more physical Coulomb gauge, and then it can be easily proved that

$$G \xrightarrow{|x| \rightarrow \infty} 0,$$

i.e., there are no asymptotic fermions.¹⁵

We can summarize this work stating that our path-integral approach has led to the exact solution of the Thirring and Schwinger models in the sense that any Green's function can be computed in a closed form. In both models gravitation does not alter the relevant properties of the flat-space-time solutions. In particular, we have shown the factorization of the (free) fermions from the bosonic sector, as was suggested in Ref. 16 for the Schwinger model in mock-Schwarzschild metric.

For the self-interacting Thirring model, our results coincide with those obtained by Birrel and Davies⁸ by extending the Klaiber¹⁴ operator solution to curved space-time. Concerning the Schwinger model, the solution is not as simple but one can discover relevant features from

our results. In particular, we prove that fermions are free at short distances for arbitrary differentiable metric. Concerning the long-distance behavior, one can infer from the correlation functions that there are no asymptotic fermion states for asymptotically flat metrics. That is, the screening phenomenon takes place. Apart from the fermions, the physical spectrum contains a massive boson with propagator defined by Eq. (54). Since this equation is

metric dependent, it cannot be solved analytically in the general case. The study of this and related problems deserves future work.

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